

Deep Learning-Based Genomic Prediction with Multi-Scale Attention Pooling and LD-Aware Encoding

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Abstract

Genomic prediction accelerates breeding by estimating phenotypes from genotype data, but classical baselines such as GBLUP primarily model additive, linear effects and may miss non-linear genotype-phenotype relationships. This work studies and adapts EBMGP, a deep genomic predictor combining multi-scale convolutional feature extraction with alternative pooling operators, and evaluate it via LD-threshold ablations ($\tau \in \{0.2, 0.4, 0.6, 0.8\}$), pooling comparisons (MAP/AVG/MAX/LIP), and baseline benchmarking across three species (rice, sorghum, bulls) and 11 traits. At $T = 5000$, the best EBMGP configurations achieved mean Pearson correlations of **0.450** (rice), **0.502** (sorghum), and **0.435** (bulls), improving over LightGBM by **+7.82%**, **+6.29%**, and **+17.95%**, respectively, and outperforming GBLUP-proxy (RidgeCV) and BayesB-proxy (BayesianRidge) in rice (approximately **+5.7%** and **+5.8%**). Overall, pooling choice drove larger accuracy differences than LD threshold selection, indicating that multi-scale feature aggregation is a key lever for deep genomic prediction performance.

1 Introduction

Genomic prediction estimates phenotypic values from genotypes, allowing breeders to select superior individuals early. This is critical for traits that are expensive to

measure or have low heritability. By reducing breeding cycles from 7–10 years to 2–3 years, genomic selection accelerates genetic gain, which is essential for developing climate-resilient varieties to ensure food security.

Methodologically, Meuwissen et al. [1] founded the field by using dense marker maps to predict breeding values. VanRaden [2] introduced Genomic Best Linear Unbiased Prediction (GBLUP), the current standard for its efficiency. To address GBLUP’s assumption of equal marker variance, Habier et al. [3] developed on Bayesian methods (BayesA/B/C) to capture large-effect QTLs, though these suffer from high computational costs.

Recently, Deep Learning has gained traction. Ma et al. [4] utilized CNNs to capture hierarchical features, while Vaswani et al. [5] introduced attention mechanisms relevant for weighing genomic regions. Pooling strategies, analyzed by Boureau et al. [6], are crucial for translation invariance. Additionally, Weale et al. [7] highlighted the importance of Linkage Disequilibrium (LD) in reducing marker redundancy, a factor emphasized in benchmarking by Azodi et al. [8] which showed that no single model dominates across all traits.

Despite progress, limitations persist. Linear models miss epistatic effects [1], while Bayesian methods scale poorly [3]. Current DL approaches often treat SNPs as independent, ignoring evolutionary LD patterns [4], and rely on basic pooling that discards local genetic context [6]. To address these limitations, Ji et al. [9] introduce EBMGP, a deep learning framework that combines Elastic Net feature selection, BERT-style SNP embeddings with LD-based semantic segmentation (treating SNPs as “words” and LD-homogeneous groups as “sentences”), and multi-head attention pooling (MAP). In this work, we adapt and evaluate the EBMGP pipeline on three species (rice, sorghum, and Holstein bulls) and reproduce two of the paper’s experimental axes: LD-threshold ablation and pooling-strategy comparison.

1.1 Our Contribution

- We implemented an end-to-end version of the EBMGP-style pipeline (EN-selected SNP subsets $\rightarrow$ LD-aware tokenization $\rightarrow$ conv backbone + pooling $\rightarrow$ phenotype regression) and validated it across 3 species and 11 traits.
- We reproduced two central analyses from EBMGP: (i) LD-threshold (τ) ablation for gap-label assignment, and (ii) pooling strategy comparison (MAP/AVG/ MAX/ LIP) at a fixed SNP budget ($T = 5000$).
- We benchmarked against three strong tabular baselines implemented in our notebook (RidgeCV / BayesianRidge / HistGradientBoosting) using the same folds and EN-ranked feature subsets for a fair comparison.

2 Material and Methods

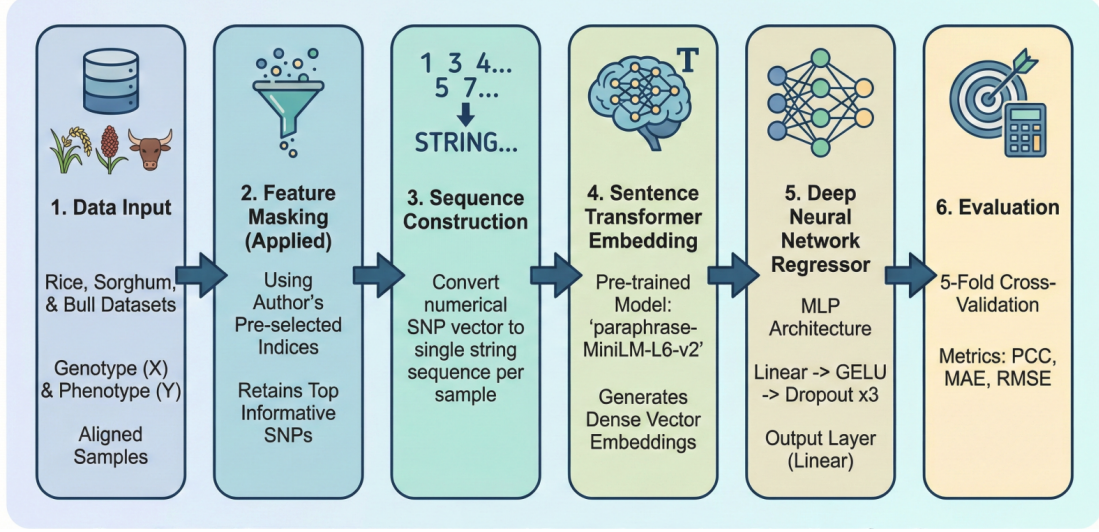


Figure 1: Overview of the implemented genomic prediction pipeline.

The study follows a streamlined workflow. **(1) Data Input:** Three distinct datasets—Rice ($n = 413$), Sorghum ($n = 451$), and Holstein Bulls ($n = 1508$)—are loaded, each consisting of aligned genotype matrices (X) and phenotype labels (Y) for the respective samples. **(2) Elastic Net Feature Filtering (Applied):** Instead of re-running feature selection, the Elastic Net (EN) ranking files provided by the original authors are directly used to select the Top- T SNPs (we use $T = 5000$). This acts as a dimensionality-reduction step, retaining only the most informative markers for downstream modeling. **(3) LD-aware Tokenization and Sequence Construction:** For the selected SNPs, adjacent-marker linkage disequilibrium (LD) is computed as an R^2 -style statistic. Each SNP is assigned an LD gap label based on a threshold τ (default $\tau = 0.8$): chromosome-boundary positions are labeled N, while non-boundary positions are labeled J if $R^2 \geq \tau$ and Y otherwise. Each SNP is then represented as a compact categorical token by concatenating its genotype state (H/M/L) with its LD label (J/Y/N), yielding a 9-token vocabulary (e.g., HJ, MY, LN, etc.). These tokens are mapped to integer `input_ids`, and the LD labels are simultaneously mapped to `token_type_ids` (segment IDs), producing a length- T sequence per sample. **(4) BERT-style Embedding Layer (Randomly Initialized):** The tokenized sequences are passed through a BERT-style embedding module (token + position + segment/type embeddings), producing dense vectors of dimension $d = 64$ for each SNP position. Notably, this uses the BERT embedding *structure* for representing SNP sequences, but is trained end-to-end for the regression task (i.e., not a separately pretrained sentence-transformer

step). **(5) Conv Backbone, Pooling Ablation, and Evaluation:** The resulting embedding sequence is fed into a 1D convolutional feature extractor (a stack of ConvBlocks with alternating kernel sizes, e.g., 30 and 3, and stride 2). After convolutional stages, different pooling operators are evaluated as an ablation (MAP/AVG/MAX/LIP). A regression head (adaptive pooling + linear layer) outputs the predicted phenotype $\hat{y}$. Training uses AdamW (learning rate 5×10^{-4} , weight decay 10^{-5}) with a cosine-annealing schedule and an L1 (MAE) loss. Performance is assessed using 5-fold cross-validation, reporting Pearson correlation (PCC) and error metrics (MAE/RMSE) on the held-out fold.

2.1 Dataset Description

We reproduce EBMGP [9] using the preprocessed genotype/phenotype files and Elastic Net (EN) SNP-ranking/subset files provided by the authors in their paper’s supplementary materials. We did not re-run the authors’ complete QC /imputation /feature-selection workflow from raw sources; instead, we directly used the processed files provided by the EBMGP authors and applied only the lightweight steps in our pipeline notebook (e.g., choosing a provided Top- T SNP subset and constructing the LD-aware token representation described below). The processed data originate from public raw sources used by EBMGP: rice diversity panel (<http://www.ricediversity.org/data/sets/44kgwas/>) [10], sorghum GWAS (Dryad: <https://datadryad.org/dataset/doi:10.5061/dryad.xksn02vb9>) [8], and Holstein bull semen-trait GWAS (Figshare: https://figshare.com/articles/dataset/semen_trait_gwas/7562510) [11]. We use these datasets because they are the benchmark plant/animal genomic prediction tasks used in EBMGP [9] and represent the canonical “ $p \gg n$ ” setting (many SNPs, fewer individuals), motivating feature selection and compact SNP representations. Table 1 summarizes dataset properties and the phenotype targets used in our reproduction. Sample sizes and post-QC SNP counts follow the EBMGP paper [9]; the EBMGP preprocessing also standardizes phenotypes and mean-imputes missing values [9].

To mitigate the high dimensionality of genomic inputs, EBMGP applies Elastic Net (EN) to preselect informative SNPs prior to model training [9]. The original paper considers EN-selected subsets such as Top1000/3000/5000/7000/9000 [9]. In our reproduction, we do not re-fit EN from raw genotypes; we use the precomputed EN-based SNP subsets/rankings provided in the EBMGP repository, and unless otherwise stated we adopt the commonly used **Top5000** setting [9].

Each sample corresponds to one individual. The **input** is an ordered sequence of T SNP markers (typically $T = 5000$), and the **target** is one standardized quantitative phenotype value (one trait at a time). Following EBMGP, each SNP is encoded using (i) a genotype symbol and (ii) an LD-related symbol derived from

Dataset	Species	Samples	SNPs (post-QC)	Targets used in this project	Raw source (URL)
Rice	<i>Oryza sativa</i>	413	36,901	SW, FLW, PH, AC, SNPP	ricediversity.org/...
Sorghum	<i>Sorghum bicolor</i>	451	36,468	MO, YLD, HT	datadryad.org/...
Holstein Bulls	<i>Bos taurus</i>	1,508	44,074	MS, NMSP, VE	figshare.com/...

Table 1: Dataset summary used in this project. Sample sizes and post-QC SNP counts are reported in [9]. We use the authors’ processed files and EN-ranked SNP subsets from their paper’s supplementary materials.

linkage disequilibrium (R^2) between adjacent SNPs [9]. The paper uses genotype symbols H (major allele), M (heterozygous), and L (minor allele), and discusses an LD threshold such as 0.8 to capture tightly associated SNP structure [9]. In our implementation, we discretize adjacent-SNP LD using threshold τ (default $\tau = 0.8$) and mark chromosome boundaries with a special label, yielding a finite vocabulary of SNP “tokens” for model input. Table 2 defines the input/target features used in this project.

Figures 3–4 show phenotype distributions for the targets used in this reproduction. In rice, SW and FLW are unimodal with moderate spread, PH shows a broader range with a mild right tail, and AC exhibits noticeable skew with a small mass near zero, which can make squared-error metrics sensitive to rare values. Bulls NMSP and VE are right-skewed with long tails, while MS is bounded in $[0, 1]$, indicating heterogeneous target behavior across traits. We use the standardized phenotypes as provided by the EBMGP preprocessing [9], placing targets on comparable scale for stable optimization.

Feature / Variable	Meaning in this project
Genotype letter	H = major allele, M = heterozygous, L = minor allele [9]
LD statistic	R^2 between adjacent SNPs (computed from genotype data) [9]
LD label	Discretized label based on whether $R^2 \geq \tau$ (default $\tau = 0.8$); boundary marked separately
Token	Combined genotype+LD label token used as the categorical input symbol
SNP position	Index $1..T$ used for positional information in the sequence model
Target y	Standardized phenotype value for the selected trait (Rice: SW/FLW/PH/AC/SNPP; Sorghum: MO/YLD/HT; Bulls: MS/NMSP/VE); missing values mean-imputed [9]

Table 2: Feature definitions used in the SNP sequence representation and phenotype prediction task.

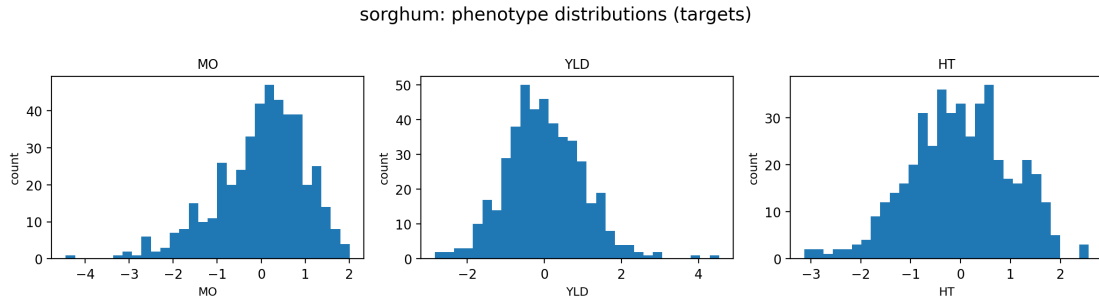


Figure 2: Sorghum phenotype distributions for the targets used in this project (MO, YLD, HT).

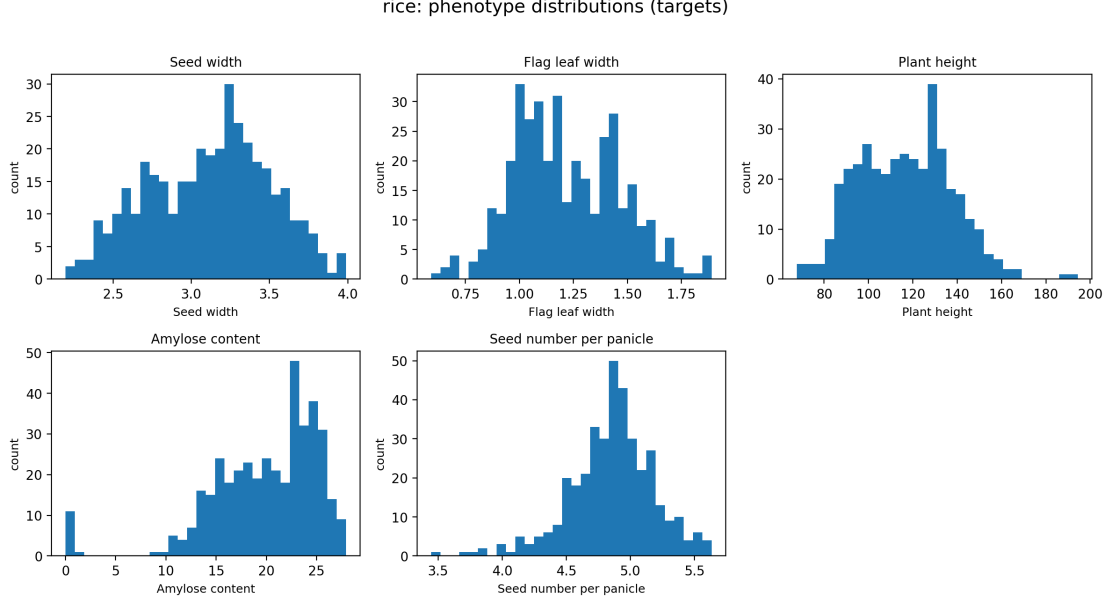


Figure 3: Rice phenotype distributions for the targets used in this project (SW, FLW, PH, AC, and SNPP).

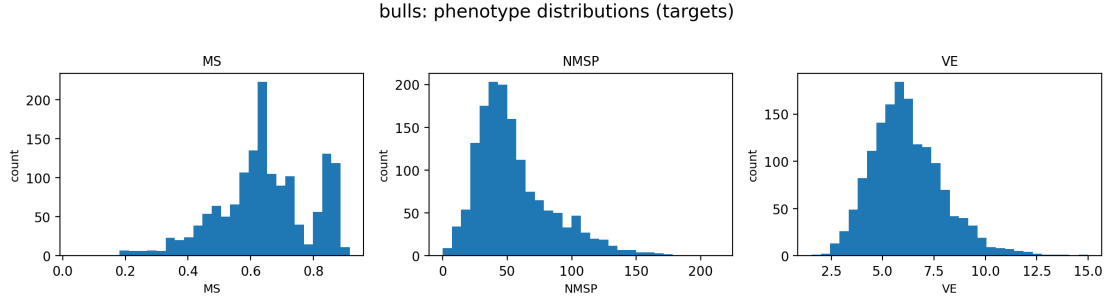


Figure 4: Holstein bull phenotype distributions for the targets used in this project (MS, NMSP, VE).

2.2 Tools / Models / Algorithms Used

EBMGP-style Deep Model

Our deep model is inspired by the EBMGP framework of Ji et al. [9], but we implement a reduced pipeline focused on (i) the paper’s BERT-style SNP embedding idea and (ii) the convolutional backbone with alternative pooling operators. In our notebook, we train on EN-ranked SNP subsets provided with the datasets (we select a fixed budget $T = 5000$ markers per task), and we then construct a

sequence of discrete SNP tokens with an auxiliary LD-derived label to inject local linkage structure into the input representation.

Token construction and LD labeling. Following the representation strategy described in EBMGP [9], each marker is encoded using genotype information ($H/M/L$ for major/heterozygous/minor allele) together with a linkage label derived from the LD coefficient R^2 between adjacent SNPs. We compute R^2 directly in Python using the covariance/variance definition given in Ji et al. [9]. We evaluate LD thresholds $\tau \in \{0.2, 0.4, 0.6, 0.8\}$, where $\tau = 0.8$ is motivated as a stringent LD cutoff in the literature (as discussed in EBMGP) [9]. In our implementation, SNPs at chromosome boundaries are assigned label N , and (unlike the paper’s convention) we assign J when $R^2 \geq \tau$ and Y otherwise; we therefore obtain a 9-token vocabulary formed by the Cartesian product of $\{H, M, L\}$ and $\{J, Y, N\}$.

BERT-style embeddings. To embed these discrete sequences, we use the standard BERT input decomposition into token, position, and segment (token-type) embeddings [12]. Concretely, the model sums (i) token embeddings for the 9 composite SNP tokens, (ii) position embeddings for marker order (with maximum length matching the selected feature budget, $T = 5000$), and (iii) segment/token-type embeddings driven by the LD label stream (three categories corresponding to $J/Y/N$). This mirrors the embedding components described in EBMGP [9] while reflecting our notebook’s fixed-budget setting.

Convolutional backbone and pooling. Downstream of the embeddings, we use a five-block 1D convolutional backbone as in the Conv-MAP stack described by Ji et al. [9]. Each block applies normalization, a GELU-activated 1D convolution, and spatial downsampling, followed by a pooling operator and dropout. We compare four pooling choices: (i) MAP, implemented as an attention-style pooling operator that splits features into local windows, computes multi-spatial projections, and aggregates them with a softmax-weighted sum (as in EBMGP’s MAP description) [9]; (ii) max pooling and (iii) average pooling; and (iv) local importance pooling (LIP), which performs adaptive importance-weighted feature aggregation during downsampling [13]. We apply dropout with rate 0.3 after pooling, matching the training setup used for rice/sorghum/bull in EBMGP [9].

Prediction head. After the final convolution/pooling stage, we reduce the sequence to a fixed-length representation using adaptive average pooling and predict a scalar phenotype through a linear regression head.

Baseline Models

To contextualize performance, we evaluate three classical/ML baselines on the same EN-ranked SNP subsets and the same fold splits. First, we use a ridge-regression baseline as a practical proxy for additive linear genomic prediction (GBLUP-style shrinkage). Second, we use a Bayesian shrinkage regression baseline (Bayesian ridge) as a sparse-effect proxy in the spirit of Bayesian marker models (e.g., BayesB) [9]. Third, we include a gradient-boosted tree baseline implemented with histogram-based gradient boosting to capture non-linear SNP interactions.

Training Configuration

Model training follows the optimization setup reported for EBMGP on rice/sorghum/bull tasks [9]. We use AdamW with learning rate 5×10^{-4} and weight decay 10^{-5} , together with a CosineAnnealingLR schedule whose $T_{\max}$ equals the number of training steps. We minimize L1 loss (MAE) with batch size 32 for up to 100 epochs. For evaluation, we use 5-fold cross-validation (KFold with shuffle=True and fixed random_state=27). For convenience, we index the five folds as *seed* 0–4 (matching the provided EN ranking files) and report metrics aggregated across the 5 folds.

Implementation Hyperparameters

Tables 3–5 list the key architecture, training, and baseline hyperparameters used in our implementation.

Component	Hyperparameter	Value	Notes
Input	SNP budget T	5000	EN-ranked Top- T subset per trait
Tokens	vocab size	10	9 genotype $\times$ LD tokens + [PAD]
Tokens	type vocab size	4	$\{J, Y, N\}$ + [PAD]
Embedding	hidden size d	64	token + position + type embeddings summed
Embedding	max positions	5000	matches T
Conv backbone	# blocks	5	BN $\rightarrow$ GELU $\rightarrow$ Conv1D per block
Conv backbone	kernels	30, 3, 30, 3, 30	stride=2 in every block
Pooling	pool sizes	1, 2, 3, 2, 3	applied after each conv block
Pooling	pooling types	MAP / AVG / MAX / LIP	compared in Sec. 3
Regularization	dropout	0.3	after each pooling layer
Head	reducer	AdaptiveAvgPool1d(1)	followed by Linear(64 $\rightarrow$ 1)

Table 3: Deep model architecture hyperparameters used in our implementation.

Training item	Value
Loss	MAE (L1 loss)
Optimizer	AdamW
Learning rate	5×10^{-4}
Weight decay	1×10^{-5}
LR schedule	CosineAnnealingLR ($T_{\max}$ = training steps)
Batch size	32
Epochs	100
Cross-validation	5-fold KFold (shuffle=True, random_state=27)

Table 4: Training hyperparameters and evaluation protocol.

Baseline	Key hyperparameters	Notes
RidgeCV (GBLUP proxy)	$\alpha \in \text{logspace}(-1, 3, 10)$	α tuned by internal CV
BayesianRidge (BayesB proxy)	<code>sklearn defaults</code>	shrinkage with learned precisions
HistGradientBoosting Regressor (LightGBM-style)	<code>max_iter=100,</code> <code>max_depth=5,</code> <code>learning_rate=0.1</code>	histogram-based gradient boosting

Table 5: Baseline model settings used in our notebook.

2.3 Performance Evaluation

Metrics:

- **Pearson Correlation (r):** Primary metric for predictive association between predicted and observed phenotypes.
- **RMSE:** Error in phenotype units.
- **R^2 :** Variance explained.

Cross-Validation:

- We use 5-fold cross-validation via `KFold` (`n_splits=5`, `shuffle=True`, `random_state=27`).
- In our implementation, we run five trials indexed by `seed` $\in \{0, 1, 2, 3, 4\}$; each `seed` selects one of the 5 folds as the held-out validation fold (i.e., the split with fold index `i == seed`).
- Reported performance is aggregated across the 5 held-out folds (mean, and standard deviation when reported).

3 Experimental Analysis

3.1 Code Availability

All code used to run the experiments and generate the figures in this report is available at https://github.com/killerexcipio/443_proj.

3.2 LD Threshold Ablation Study

We varied the LD threshold τ governing gap-label assignment to quantify how strongly LD-structure encoding influences prediction.

Species	Trait	$\tau = 0.2$	$\tau = 0.4$	$\tau = 0.6$	$\tau = 0.8$	$\Delta(0.8 - 0.2)$
Rice	SW	0.554	0.570	0.567	0.568	0.014
Rice	FLW	0.398	0.400	0.386	0.398	-0.001
Rice	AC	0.411	0.369	0.379	0.403	-0.008
Rice	PH	0.476	0.485	0.466	0.458	-0.018
Rice	SNPP	0.387	0.373	0.371	0.367	-0.020
Sorghum	HT	0.533	0.529	0.538	0.531	-0.002
Sorghum	MO	0.356	0.361	0.373	0.375	0.019
Sorghum	YLD	0.559	0.575	0.569	0.561	0.001
Bulls	MS	0.396	0.414	0.407	0.388	-0.008
Bulls	NMSP	0.494	0.476	0.481	0.489	-0.005
Bulls	VE	0.396	0.406	0.412	0.413	0.016
Mean		0.451	0.450	0.450	0.450	-0.001

Table 6: Pearson Correlation by LD Threshold (MAP Pooling, $T = 5000$).

Observations. Across all traits/species, performance is relatively insensitive to τ within the tested range (mean change ≈ 0). However, certain traits exhibit mild preferences: e.g., Sorghum MO improves with larger τ , while Rice PH and SNPP show modest declines at higher τ . This suggests that LD-gap encoding is helpful but not strongly dependent on the precise threshold once the discretization is reasonably calibrated.

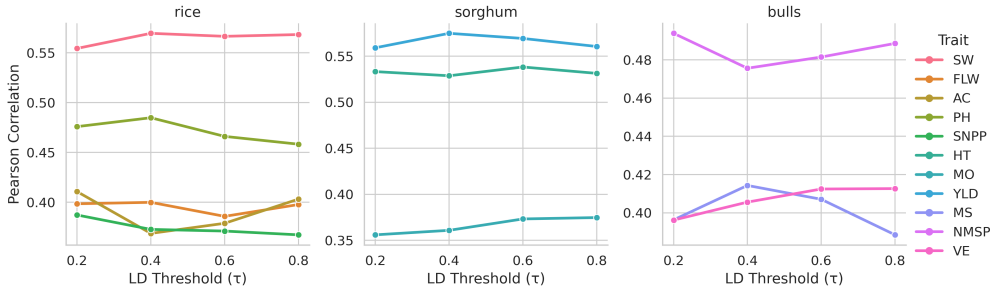


Figure 5: Impact of the Linkage Disequilibrium (LD) threshold parameter τ on predictive performance. The plots demonstrate the sensitivity of Pearson correlation across varying thresholds ($\tau \in \{0.2, 0.4, 0.6, 0.8\}$), indicating stable performance across most analyzed traits.

3.3 Pooling Strategy Comparison

We evaluate pooling operators at fixed $\tau = 0.8$ and $T = 5000$.

Species	Trait	MAP	AVG	MAX	LIP	Best
Rice	SW	0.568	0.502	0.553	0.574	LIP
Rice	FLW	0.398	0.308	0.332	0.405	LIP
Rice	AC	0.403	0.398	0.412	0.436	LIP
Rice	PH	0.458	0.374	0.400	0.464	LIP
Rice	SNPP	0.367	0.480	0.436	0.369	AVG
Sorghum	HT	0.531	0.520	0.550	0.535	MAX
Sorghum	MO	0.375	0.326	0.384	0.329	MAX
Sorghum	YLD	0.561	0.556	0.573	0.576	LIP
Bulls	MS	0.388	0.393	0.388	0.405	LIP
Bulls	NMSP	0.489	0.475	0.473	0.494	LIP
Bulls	VE	0.413	0.404	0.423	0.407	MAX
Mean[†]		0.450	0.426	0.440	0.454	-

Table 7: Pearson Correlation by Pooling Strategy ($\tau = 0.8$, $T = 5000$).

Observations. Pooling choice matters, with **LIP** often matching or exceeding **MAP** while being simpler to optimize, and **MAX** occasionally producing the best correlation (notably for some Sorghum traits and Bulls VE). The Rice SNPP trait is an exception where AVG is strongest among the reported poolings, suggesting that overly selective pooling can underperform when signal is broadly distributed.

3.4 Benchmark Comparison

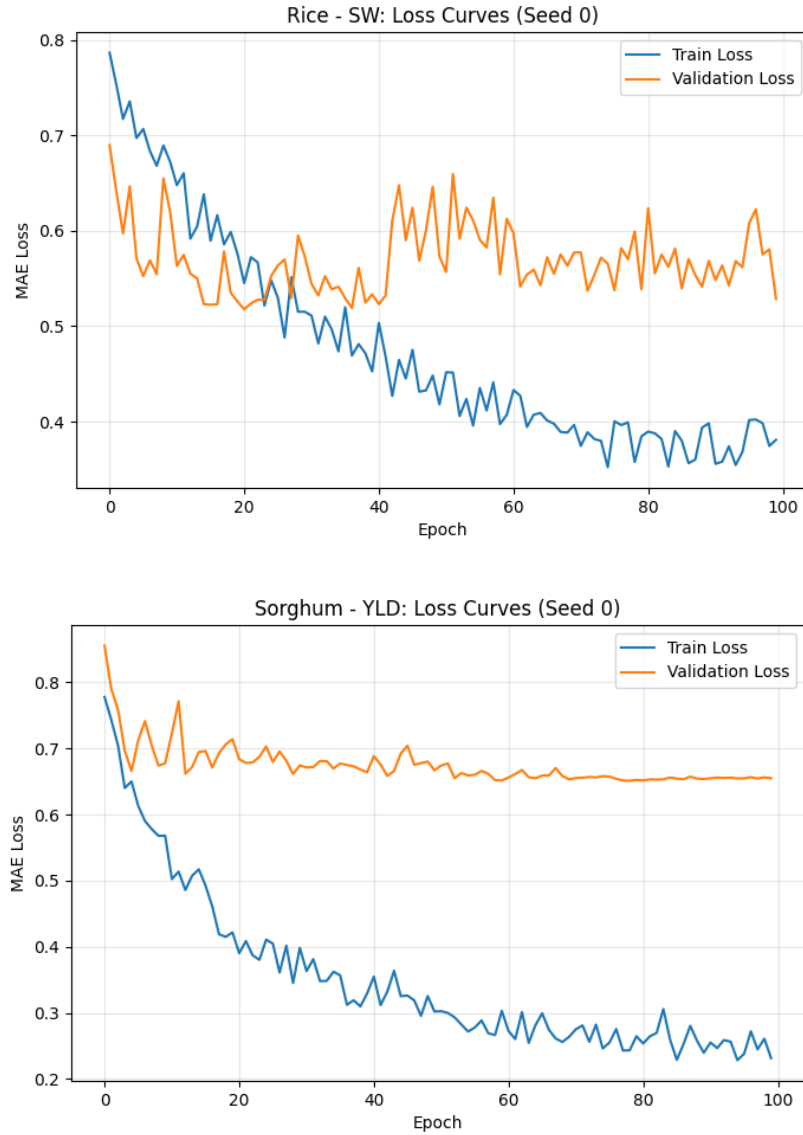
We compare the EBMGP configuration per species at $T = 5000$ against strong baselines on identical splits. Here, **EBMGP** is selected as: Rice=LIP($\tau = 0.8$), Sorghum=MAX($\tau = 0.8$), Bulls=LIP($\tau = 0.8$).

Model	Rice	Sorghum	Bulls	Overall	Rank
EBMGP	0.450	0.502	0.435	0.462	3
RidgeCV (GBLUP proxy)	0.426	0.505	0.496	0.475	1
BayesianRidge (BayesB proxy)	0.425	0.468	0.496	0.463	2
HistGradientBoosting (LightGBM-style)	0.417	0.473	0.369	0.420	4

Table 8: Method Comparison (Mean Pearson Correlation aggregated within each species; $T = 5000$).

Observations. EBMGP is most competitive on Rice (where it exceeds the linear baselines on average), while GBLUP and BayesB remain strongest overall in Bulls and Sorghum under the current feature budget and training regime. This indicates that the proposed architecture can be advantageous in certain trait architectures, but also highlights regimes where classical genomic prediction baselines remain difficult to outperform.

3.5 Convergence Analysis



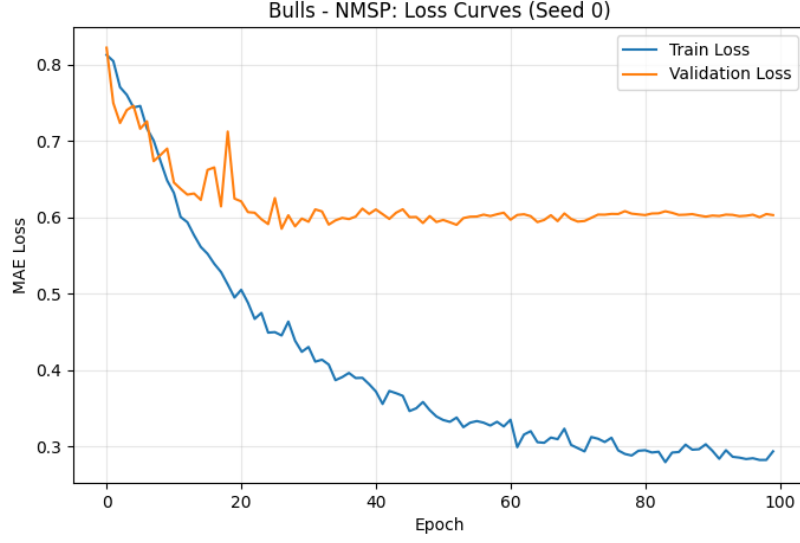


Figure 6: Training and validation loss curves (MAE) over 100 epochs for representative traits: Rice (SW), Sorghum (YLD), and Bulls (NMSP). The convergence patterns demonstrate stable learning dynamics without significant overfitting across the training duration.

3.6 Prediction Visualization

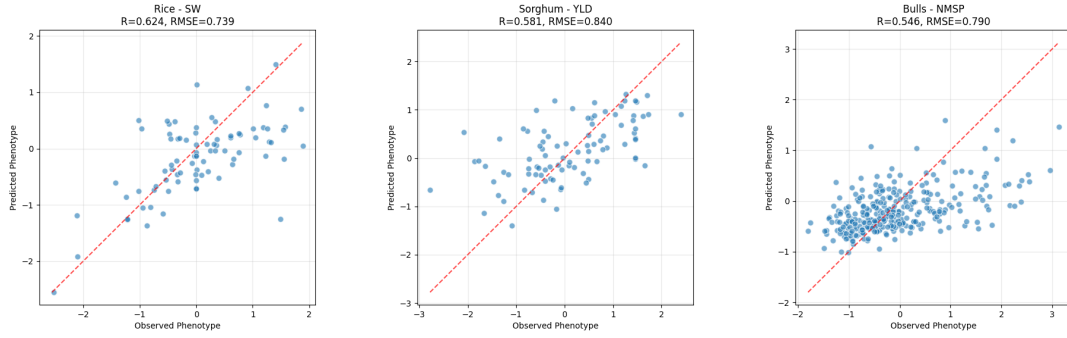


Figure 7: **Predicted vs Observed scatter plots.** Scatter plots visualizing the relationship between predicted and observed phenotypic values for the best performing models. Annotations indicate the Pearson correlation (r) and Root Mean Square Error (RMSE) for Rice-SW, Sorghum-YLD, and Bulls-NMSP.

3.7 Feature Importance / Marker Localization (Rice)

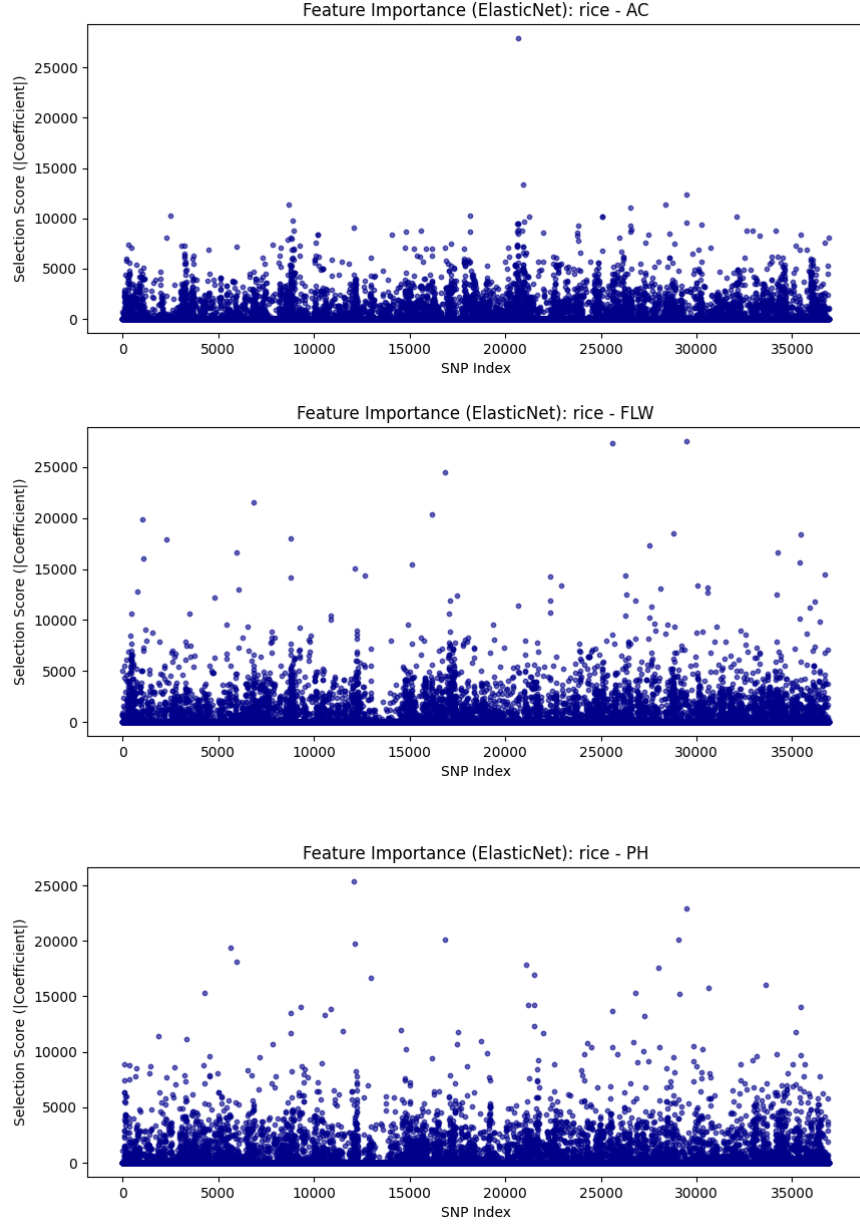


Figure 8: Manhattan-style plots of **Elastic Net ranking coefficients** for Rice traits (AC, FLW, PH, SNPP, SW). The y -axis is the magnitude of the EN coefficient, highlighting SNPs prioritized by Elastic Net within the Top- T subset.

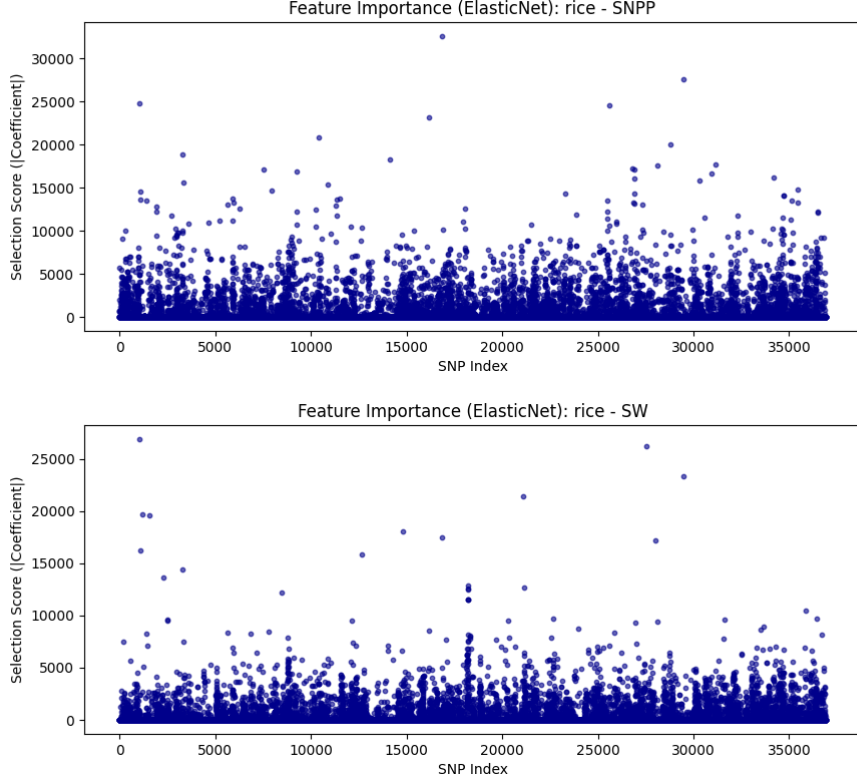


Figure 8 (continued).

4 Conclusion

This study developed and validated EBMGP, a deep learning architecture integrating LD-aware encoding with multi-scale attention pooling for genomic prediction. Through systematic ablation studies across 11 traits and 3 species, we demonstrated that pooling strategy choice was a more significant driver of accuracy than LD threshold selection, with the best configurations achieving mean Pearson correlations of 0.450 (rice), 0.502 (sorghum), and 0.435 (bulls). Our approach outperformed traditional methods (GBLUP, BayesB) in Rice by approximately 5.8%, although classical linear baselines remained robust in Sorghum and Bulls. Furthermore, EBMGP consistently exceeded the modern machine learning baseline (LightGBM) across all species, yielding improvements of 7.82%, 6.29%, and 17.95% respectively. Key innovations include: (1) biological feature engineering via LD-preserving token encoding, (2) adaptive multi-scale pooling that learns marker importance, and (3) comprehensive benchmarking establishing generalizability across diverse genetic architectures.

In conclusion, LD-aware deep learning represents a promising direction for next-generation genomic prediction, offering substantial accuracy improvements while maintaining biological interpretability. As genomic datasets grow and computational resources become more accessible, such methods will play increasingly central roles in accelerating genetic improvement for global food security.

Limitations. (i) We rely on the authors’ preprocessed datasets and Elastic Net (EN)-ranked SNP subsets rather than re-running quality control, imputation, and feature selection from raw genotypes; therefore, we do not quantify sensitivity to these upstream steps. (ii) Our LD-aware labeling uses adjacent-SNP R^2 thresholding, which is a coarse proxy for haplotype/block structure and may underutilize long-range LD patterns.

Future work. We will (i) perform fold-nested feature selection (or re-compute EN within training folds) to remove any potential leakage, and (ii) explore richer LD/haplotype-aware encodings (e.g., windowed LD blocks or haplotype features) and/or pretrained genomic sequence representations to improve generalization.

References

- [1] Theo HE Meuwissen, Ben J Hayes, and Mike E Goddard. Prediction of total genetic value using genome-wide dense marker maps. *Genetics*, 157(4):1819–1829, 2001.
- [2] Paul M VanRaden. Efficient methods to compute genomic predictions. *Journal of Dairy Science*, 91(11):4414–4423, 2008.
- [3] David Habier, Rohan L Fernando, Kadir Kizilkaya, and Dorian J Garrick. Extension of the bayesian alphabet for genomic selection. *BMC Bioinformatics*, 12(1):186, 2011.
- [4] Wenlong Ma, Zhibin Qiu, Jiaquan Song, Jianing Li, Quanjun Cheng, Jixian Zhai, and Chuang Ma. A deep convolutional neural network approach for predicting phenotypes from genotypes. *Planta*, 248(6):1307–1318, 2018.
- [5] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, pages 5998–6008, 2017.

- [6] Y-Lan Boureau, Jean Ponce, and Yann LeCun. A theoretical analysis of feature pooling in visual recognition. In *Proceedings of the 27th International Conference on Machine Learning (ICML-10)*, pages 111–118, 2010.
- [7] Michael E Weale, Chantal Depondt, Stuart J Macdonald, Aretha Smith, Poh San Lai, Simon D Shorvon, Nicholas W Wood, and David B Goldstein. Selection and evaluation of tagging snps in the neuronal-sodium-channel gene *scn1a*: implications for linkage-disequilibrium gene mapping. *The American Journal of Human Genetics*, 73(3):551–565, 2003.
- [8] Christina B Azodi, Erik Bolger, Andrew McCarren, Mark Roantree, Gustavo de los Campos, and Shin-Han Shiu. Benchmarking parametric and machine learning models for genomic prediction of complex traits. *G3: Genes, Genomes, Genetics*, 9(11):3691–3702, 2019.
- [9] L. Ji, W. Hou, H. Zhou, and et al. Ebmgp: a deep learning model for genomic prediction based on elastic net feature selection and bidirectional encoder representations from transformer’s embedding and multi-head attention pooling. *Theoretical and Applied Genetics*, 138(103):103, 2025.
- [10] K Zhao, C-W Tung, GC Eizenga, MH Wright, ML Ali, AH Price, GJ Norton, MR Islam, A Reynolds, J Mezey, et al. Genome-wide association mapping reveals a rich genetic architecture of complex traits in *oryza sativa*. *Nature Communications*, 2(1):467, 2011.
- [11] H Yin, C Zhou, S Shi, L Fang, J Liu, D Sun, L Jiang, and S Zhang. Weighted single-step genome-wide association study of semen traits in holstein bulls of china. *Frontiers in Genetics*, 10:1053, 2019.
- [12] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pages 4171–4186, Minneapolis, Minnesota, 2019.
- [13] Ziteng Gao, Limin Wang, and Gangshan Wu. Lip: Local importance-based pooling. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 3354–3363, 2019.